

Text Analysis for Social Scientists and Leaders

Assignment 3

You were introduced to some functions for calculating word embeddings. Next, you will apply it to a dataset of consumer complaints to the CFPB. Your primary outcome of interest is whether the company's response to the claim is disputed or not (binary, 0 or 1).

Before you start, filter the data to focus on only the "Credit Reporting" products. Then randomly split the CFPB data into a training set (80% of documents) and a test set (20% of documents). Also, please download the full 6GB fasttext word embedding file, don't just use the small one shared in class.

1. Train a LASSO model using ngrams from the narrative text box to predict whether each claim is disputed. Train another LASSO model using word2vec embeddings from the same text as features. Finally, train a LASSO model that uses both the ngram and embedding features together. Calculate the accuracy of these three models. (hint - the training set size is small, so you may need to set `s="lambda.min"` when you make predictions or extract coefficients from the glmnet model)

2. Choose two other benchmarks you would like to use to compare to the trained models. Create a plot showing the average accuracy scores (with 95% confidence intervals) of the models and benchmarks from questions 1 and 2. Give a description of what you found.

3. Use the Distributed Dictionary Representation technique to compute similarity of all the narratives to a dictionary - the Loughran & McDonald uncertainty dictionary - to create an "uncertainty score" for each narrative. Now notice there is a "sub-issue" column, indicating different types of issues within the credit-reporting product. Calculate the average similarity for each sub issue (mean and standard error) and put these all on a plot. Make sure the sub-issues are listed in order, from highest to lowest average uncertainty. Do these differences make sense?

4. Use these uncertainty scores to predict if the complaint is disputed or not. Then use `dfm_lookup` to apply the dictionary list the traditional way (i.e. without the vector representation). Make a plot comparing these two accuracy scores.

[note: if you can't get spacy working, then you can run this function on posit.cloud easily, if you sign up for a free account]